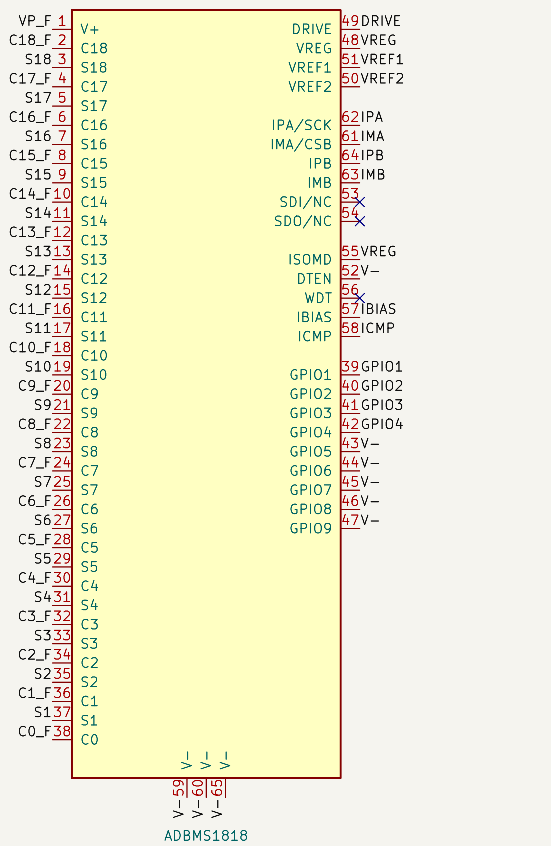
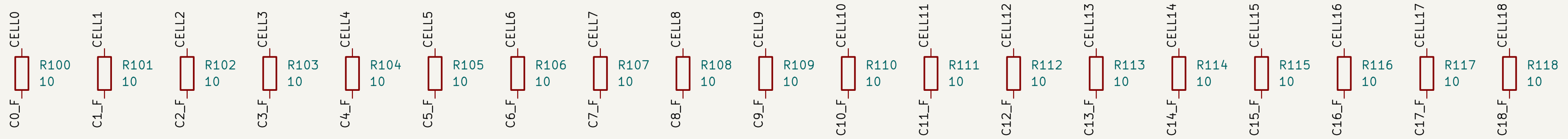
Wired with net labels: labels with the same name are one net. TAPn = connector side of fuse F1nn, CELLn = after it. V− (chip ground) has its own fuse F120.
ADBMS1818 Figure 112, isoSPI filtering per Fig. 104(B). See ../docs/slave−18s−design−basis.md.

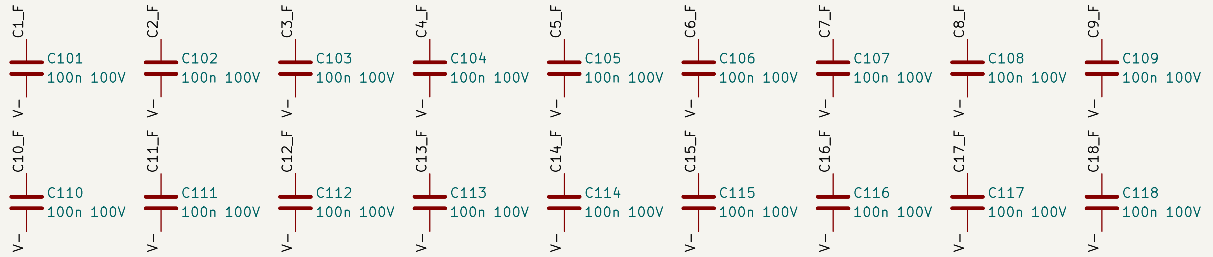
U1 ADBMS1818 (64−LQFP_EP, EP = V−)


U1
VP_F 1 V+
C18_F 2 C18
S18 3 S18
C17_F 4 C17
S17 5 S17
C16_F 6 C16
S16 7 S16
C15_F 8 C15
S15 9 S15
C14_F 10 C14
S14 11 S14
C13_F 12 C13
S13 13 S13
C12_F 14 C12
S12 15 S12
C11_F 16 C11
S11 17 S11
C10_F 18 C10
S10 19 S10
C9_F 20 C9
S9 21 S9
C8_F 22 C8
S8 23 S8
C7_F 24 C7
S7 25 S7
C6_F 26 C6
S6 27 S6
C5_F 28 C5
S5 29 S5
C4_F 30 C4
S4 31 S4
C3_F 32 C3
S3 33 S3
C2_F 34 C2
S2 35 S2
C1_F 36 C1
S1 37 S1
C0_F 38 C0
DRIVE 49 DRIVE
VREG 48 VREG
VREF1 51 VREF1
VREF2 50 VREF2
IPA/SCK 62 IPA
IMA/CSB 61 IMA
IPB 64 IPB
IMB 63 IMB
SDI/NC 53
SDO/NC 54
ISOMD 55 VREG
DTEN 52 V−
WDT 56
IBIAS 57 IBIAS
ICMP 58 ICMP
GPIO1 39 GPIO1
GPIO2 40 GPIO2
GPIO3 41 GPIO3
GPIO4 42 GPIO4
GPIO5 43 V−
GPIO6 44 V−
GPIO7 45 V−
GPIO8 46 V−
GPIO9 47 V−
V− 59 V−
V− 60 V−
V− 65 V−
ADBMS1818

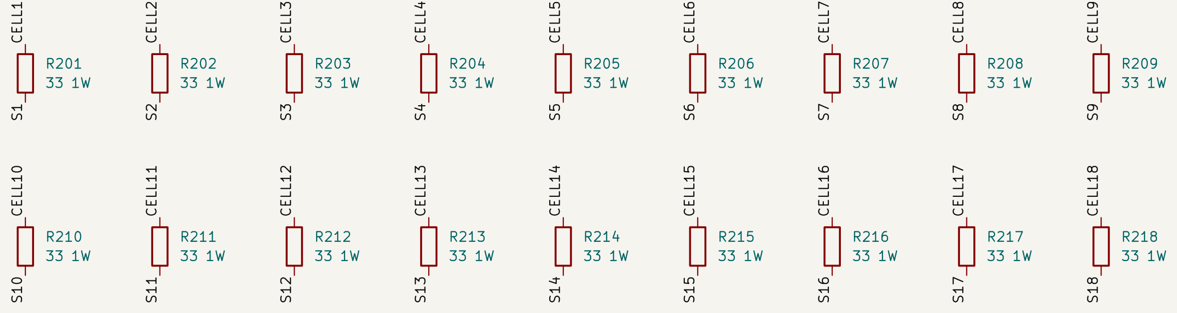
CELL INPUTS R100−R118: 10 ohm series on C0−C18


CELL0 R100 10 C0_F
CELL1 R101 10 C1_F
CELL2 R102 10 C2_F
CELL3 R103 10 C3_F
CELL4 R104 10 C4_F
CELL5 R105 10 C5_F
CELL6 R106 10 C6_F
CELL7 R107 10 C7_F
CELL8 R108 10 C8_F
CELL9 R109 10 C9_F
CELL10 R110 10 C10_F
CELL11 R111 10 C11_F
CELL12 R112 10 C12_F
CELL13 R113 10 C13_F
CELL14 R114 10 C14_F
CELL15 R115 10 C15_F
CELL16 R116 10 C16_F
CELL17 R117 10 C17_F
CELL18 R118 10 C18_F

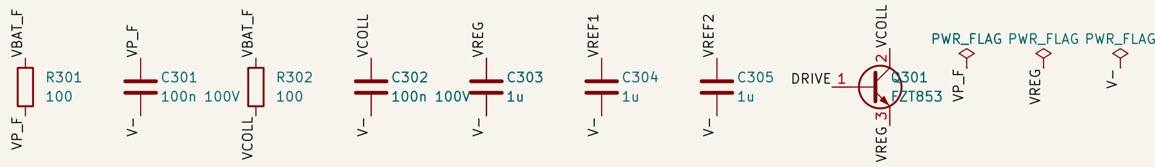
C101−C118: 100 nF 100 V, Cn (after 10 ohm) to V−


C1_F C101 100n 100V V−
C2_F C102 100n 100V V−
C3_F C103 100n 100V V−
C4_F C104 100n 100V V−
C5_F C105 100n 100V V−
C6_F C106 100n 100V V−
C7_F C107 100n 100V V−
C8_F C108 100n 100V V−
C9_F C109 100n 100V V−
C10_F C110 100n 100V V−
C11_F C111 100n 100V V−
C12_F C112 100n 100V V−
C13_F C113 100n 100V V−
C14_F C114 100n 100V V−
C15_F C115 100n 100V V−
C16_F C116 100n 100V V−
C17_F C117 100n 100V V−
C18_F C118 100n 100V V−

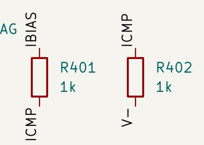
R201−R218: 33 ohm 1 W balance, cell n node (before 10 ohm) to Sn


CELL1 R201 33 1W S1
CELL2 R202 33 1W S2
CELL3 R203 33 1W S3
CELL4 R204 33 1W S4
CELL5 R205 33 1W S5
CELL6 R206 33 1W S6
CELL7 R207 33 1W S7
CELL8 R208 33 1W S8
CELL9 R209 33 1W S9
CELL10 R210 33 1W S10
CELL11 R211 33 1W S11
CELL12 R212 33 1W S12
CELL13 R213 33 1W S13
CELL14 R214 33 1W S14
CELL15 R215 33 1W S15
CELL16 R216 33 1W S16
CELL17 R217 33 1W S17
CELL18 R218 33 1W S18

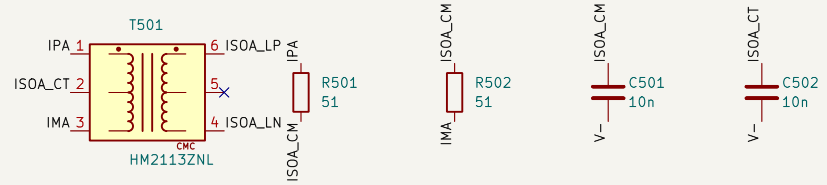
SUPPLY: V+ filter, DRIVE NPN −> VREG, VREF1/2


VBAT_F R301 100 VP_F
VP_F C301 100n 100V V−
VBAT_F R302 100 VCOLL
VCOLL C302 100n 100V V−
VREG C303 1u V−
VREF1 C304 1u V−
VREF2 C305 1u V−
DRIVE 1 Q301 FZT853 2 VCOLL 3 VREG
PWR_FLAG VP_F
PWR_FLAG VREG
PWR_FLAG V−

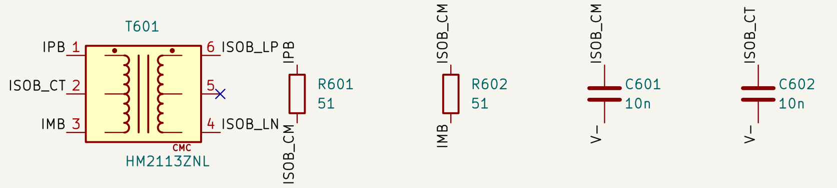
isoSPI BIAS: IBIAS−1k−ICMP−1k−V−


IBIAS R401 1k ICMP
ICMP R402 1k V−

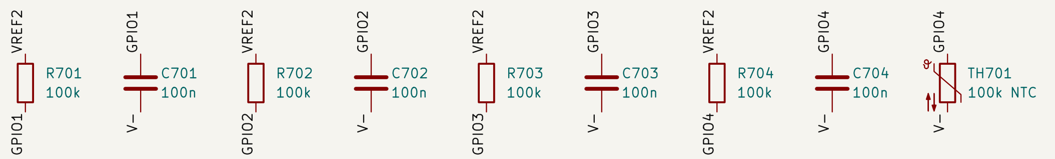
isoSPI PORT A: 2x51 ohm split term + 10 nF CTs


T501
IPA 1
ISOA_CT 2
IMA 3
6 ISOA_LP
5
4 ISOA_LN
CMC
HM2113ZNL
IPA R501 51 ISOA_CM
ISOA_CM R502 51 IMA
ISOA_CM C501 10n V−
ISOA_CT C502 10n V−

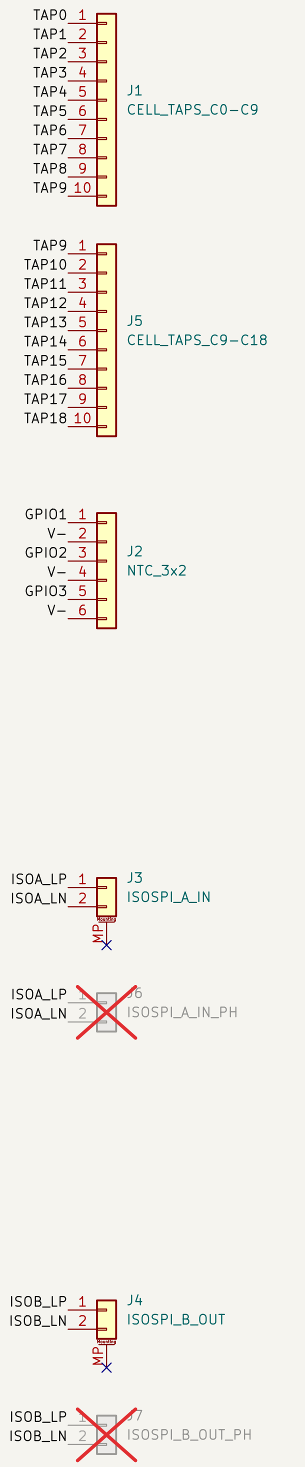
isoSPI PORT B: 2x51 ohm split term + 10 nF CTs


T601
IPB 1
ISOB_CT 2
IMB 3
6 ISOB_LP
5
4 ISOB_LN
CMC
HM2113ZNL
IPB R601 51 ISOB_CM
ISOB_CM R602 51 IMB
ISOB_CM C601 10n V−
ISOB_CT C602 10n V−

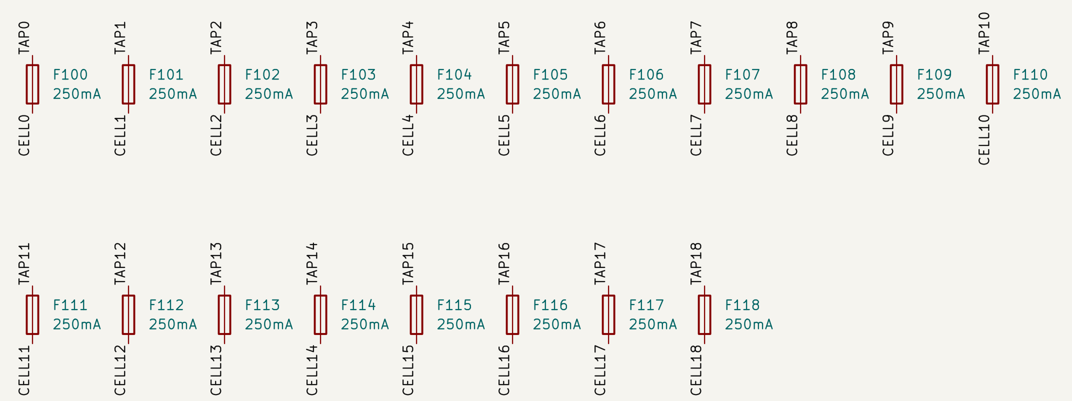
THERMISTORS: 100k to VREF2 + 100 nF, GPIO1−3 cell NTC, GPIO4 onboard


VREF2 R701 100k GPIO1
GPIO1 C701 100n V−
VREF2 R702 100k GPIO2
GPIO2 C702 100n V−
VREF2 R703 100k GPIO3
GPIO3 C703 100n V−
VREF2 R704 100k GPIO4
GPIO4 C704 100n V−
GPIO4 TH701 100k NTC V−

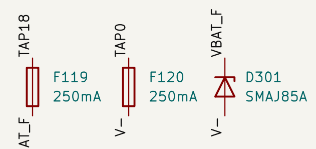
CONNECTORS (JST PH)


TAP0 1
TAP1 2
TAP2 3
TAP3 4
TAP4 5
TAP5 6
TAP6 7
TAP7 8
TAP8 9
TAP9 10
J1
CELL_TAPS_C0−C9
TAP9 1
TAP10 2
TAP11 3
TAP12 4
TAP13 5
TAP14 6
TAP15 7
TAP16 8
TAP17 9
TAP18 10
J5
CELL_TAPS_C9−C18
GPIO1 1
V− 2
GPIO2 3
V− 4
GPIO3 5
V− 6
J2
NTC_3x2
ISOA_LP 1
ISOA_LN 2
MP
Mounting
J3
ISOSPI_A_IN
ISOA_LP 1
ISOA_LN 2
J6
ISOSPI_A_IN_PH
ISOB_LP 1
ISOB_LN 2
MP
Mounting
J4
ISOSPI_B_OUT
ISOB_LP 1
ISOB_LN 2
J7
ISOSPI_B_OUT_PH

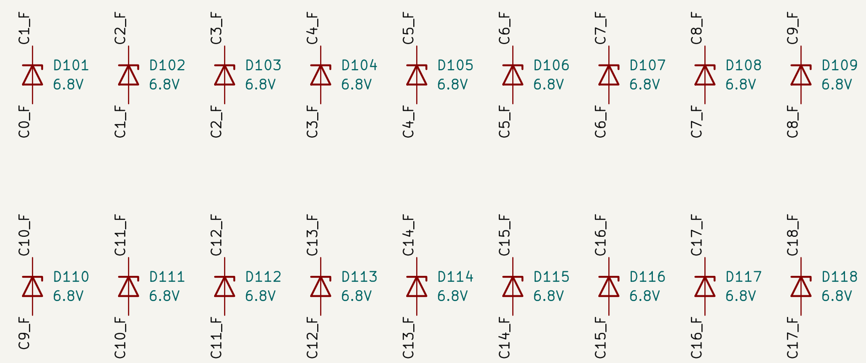
TAP FUSES F100−F118: 250 mA on every cell tap, connector side (AN−2093 Fig. 11)


TAP0 F100 250mA CELL0
TAP1 F101 250mA CELL1
TAP2 F102 250mA CELL2
TAP3 F103 250mA CELL3
TAP4 F104 250mA CELL4
TAP5 F105 250mA CELL5
TAP6 F106 250mA CELL6
TAP7 F107 250mA CELL7
TAP8 F108 250mA CELL8
TAP9 F109 250mA CELL9
TAP10 F110 250mA CELL10
TAP11 F111 250mA CELL11
TAP12 F112 250mA CELL12
TAP13 F113 250mA CELL13
TAP14 F114 250mA CELL14
TAP15 F115 250mA CELL15
TAP16 F116 250mA CELL16
TAP17 F117 250mA CELL17
TAP18 F118 250mA CELL18

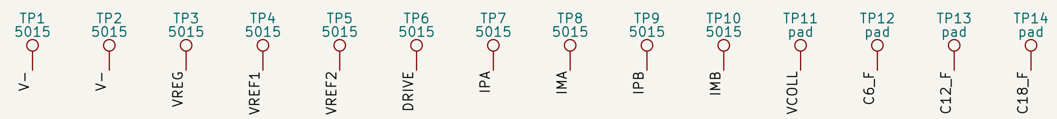
SUPPLY FUSES + TVS F119 top of stack, F120 V−, D301 85 V TVS after F119


TAP18 F119 250mA VBAT_F
TAP0 F120 250mA V−
VBAT_F D301 SMAJ85A V−

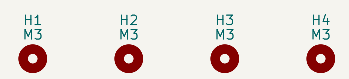
ZENER CLAMPS D101−D118: 6.8 V from Cn_F (cathode) to C(n−1)_F, abs max 8 V per pair


C1_F D101 6.8V C0_F
C2_F D102 6.8V C1_F
C3_F D103 6.8V C2_F
C4_F D104 6.8V C3_F
C5_F D105 6.8V C4_F
C6_F D106 6.8V C5_F
C7_F D107 6.8V C6_F
C8_F D108 6.8V C7_F
C9_F D109 6.8V C8_F
C10_F D110 6.8V C9_F
C11_F D111 6.8V C10_F
C12_F D112 6.8V C11_F
C13_F D113 6.8V C12_F
C14_F D114 6.8V C13_F
C15_F D115 6.8V C14_F
C16_F D116 6.8V C15_F
C17_F D117 6.8V C16_F
C18_F D118 6.8V C17_F

TEST POINTS TP1−TP10 Keystone 5015 loops, TP11−TP14 bare 1.5 mm pads (HV, no loop)


TP1 5015 V−
TP2 5015 V−
TP3 5015 VREG
TP4 5015 VREF1
TP5 5015 VREF2
TP6 5015 DRIVE
TP7 5015 IPA
TP8 5015 IMA
TP9 5015 IPB
TP10 5015 IMB
TP11 pad VCOLL
TP12 pad C6_F
TP13 pad C12_F
TP14 pad C18_F

MOUNTING HOLES H1−H4: M3, no copper, positions set with the board outline


H1 M3
H2 M3
H3 M3
H4 M3

Not for manufacture
Baseline: ADBMS1818 datasheet Figure 112 typical application

Sheet: /
File: adbms1818−slave−18s.kicad_sch

Title: ADBMS1818 18S slave module

Size: User
Date: 2026−11−06
Rev: 0−draft
KiCad E.D.A. 10.0.6
Id: 1/1